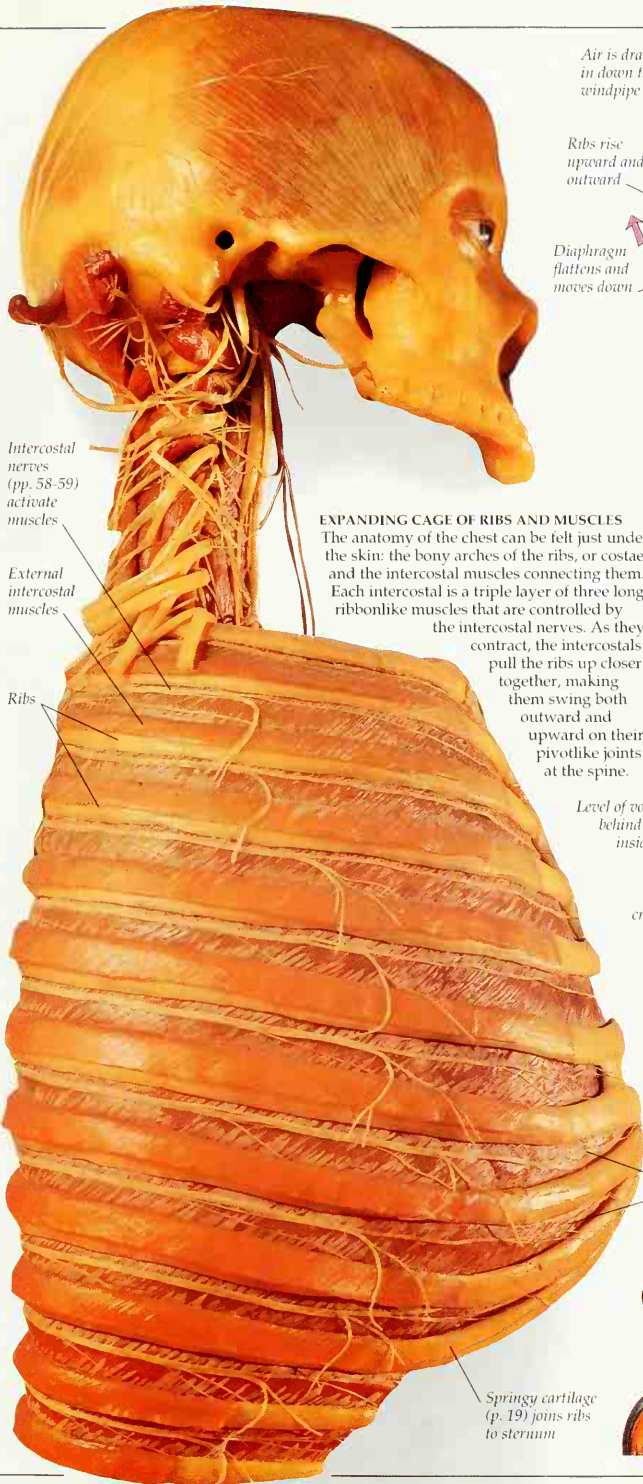
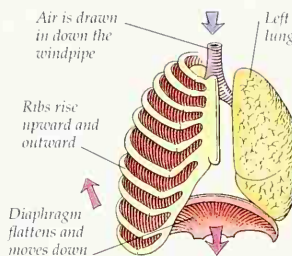




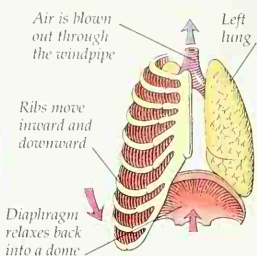
EXPANDING CAGE OF RIBS AND MUSCLES

The anatomy of the chest can be felt just under the skin: the bony arches of the ribs, or costae, and the intercostal muscles connecting them. Each intercostal is a triple layer of three long, ribbonlike muscles that are controlled by the intercostal nerves. As they contract, the intercostals pull the ribs up closer together, making them swing both outward and upward on their pivotlike joints at the spine.



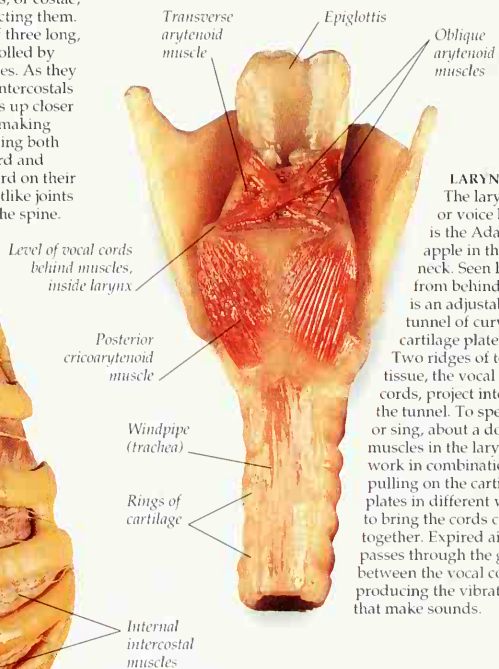
INSPIRATION

There are two main sets of muscles involved in breathing – the intercostals and the diaphragm. The intercostals contract to raise the ribs, and the diaphragm contracts from a bowl into a flatter, plate shape. This action is noticeable in gasps after strenuous exercise.



EXPIRATION

The lungs expand during inspiration because they are sealed by fluid onto the inside of the rib cage. To breathe out, both the diaphragm and intercostal muscles relax. Much like a balloon deflating, the lungs spring back to size and blow out their air.



WHAT A YAWN

When breathing is quiet and shallow, there may be a lack of airflow to the lungs. This causes a rise in the level of carbon dioxide in the blood, which is normally expelled by expiration, and a fall in the level of oxygen. This change triggers an extra deep breath, or yawn, to blow off carbon dioxide and obtain more oxygen.